UNIT 5 LEET CODE EX

1) Product Sales Analysis I

Table: Sales

Column Name	Type
sale_id	int
product_id	int
year	int
quantity	int
price	int

(sale_id, year) is the primary key of this table.product_id is a foreign key to table Product. Note that the price is per unit

Table: Product

Column Name	Type
product_id	int
product_name	varchar

product_id is the primary key of this table.

TASK:

Write an SQL query that reports the product_name, year, and price for each sale_id in the Sales table.Return the resulting table in any order.

SOLUTION:

MySQL Auto

```mysql
# Write your MySQL query statement below
select p.product_name, s.year, s.price
from Sales s
left join Product p
on s.product_id = p.product_id
```

Saved

Testcase | Test Result

Input

Sales =

sale_id	product_id	year	quantity	price
1	100	2008	10	5000
2	100	2009	12	5000
7	200	2011	15	9000

Product =

product_id	product_name
100	Nokia
200	Apple
300	Samsung

Output

product_name	year	price
Nokia	2008	5000
Nokia	2009	5000
Apple	2011	9000

2) Product Sales Analysis II

Table: Sales

Column Name	Type
sale_id	int
product_id	int
year	int
quantity	int
price	int

(sale_id, year) is the primary key of this table.product_id is a foreign key to table Product. Note that the price is per unit

Table: Product

Column Name	Type
product_id	int
product_name	varchar

product_id is the primary key of this table.

TASK:

Write an SQL query that reports the total quantity sold for every product id.

SOLUTION:

MySQL Query :

```
select product_id, sum(quantity) as total_quantity

from Sales

group by product_id;
```

Result table:

product_id	total_quantity
100	22
200	15

3) Sales Analysis III

Table: Sales

Column Name	Type
sale_id	int
product_id	int
year	int
quantity	int
price	int

(sale_id, year) is the primary key of this table.product_id is a foreign key to table Product. Note that the price is per unit.

Table: Product

Column Name	Type
product_id	int
product_name	varchar

product_id is the primary key of this table.

TASK

Write an SQL query that reports the products that were only sold in spring 2019. That is, between 2019-01-01 and 2019-03-31 inclusive.

SOLUTION:

```sql
# Write your MySQL query statement below
select p.product_id, p.product_name from Product p
left join Sales s on p.product_id = s.product_id
group by p.product_id having sum(s.sale_date between '2019-01-01' and '2019-03-31') = count(s.sale_date)
```

Saved

Testcase | Test Result

Case 1

Input

Product =

product_id	product_name	unit_price
1	S8	1000
2	G4	800
3	iPhone	1400

Sales =

seller_id	product_id	buyer_id	sale_date	quantity	price
1	1	1	2019-01-21	2	2000
1	2	2	2019-02-17	1	800
2	2	3	2019-06-02	1	800
3	3	4	2019-05-13	2	2800

Output

product_id	product_name
1	S8

4) Product Price At A Given Date

Table: Products

Column Name	Type
product_id	int
new_price	int
change_date	date

TASK:
Write a solution to find the prices of all products on the date 2019-08-16.
SOLUTION:

MySQL Auto

```mysql
# Write your MySQL query statement below
WITH LatestPrice AS (
SELECT
product_id,
new_price,
ROW_NUMBER() OVER (
PARTITION BY product_id
ORDER BY change_date DESC
) AS rn
FROM Products
WHERE change_date <= '2019-08-16'
)
SELECT
p.product_id,
COALESCE(lp.new_price, 10) AS price
FROM
(SELECT DISTINCT product_id FROM Products) p
LEFT JOIN LatestPrice lp
ON p.product_id = lp.product_id
AND lp.rn = 1;
```

Case 1

Input

Products =

product_id	new_price	change_date
1	20	2019-08-14
2	50	2019-08-14
1	30	2019-08-15
1	35	2019-08-16
2	65	2019-08-17
3	20	2019-08-18

Output

product_id	price
1	35
2	50
3	10

5) Immediate Food Delivery II

Table: Delivery

Column Name	Type
delivery_id	int
customer_id	int
order_date	date
customer_pref_delivery_date	date

TASK:

Write a solution to find the percentage of immediate orders in the first orders of all customers, rounded to 2 decimal places.

SOLUTION:

```sql
# Write your MySQL query statement below
WITH FirstOrders AS (
SELECT *,
ROW_NUMBER() OVER (
PARTITION BY customer_id
ORDER BY order_date
) AS rn
FROM Delivery
)
SELECT
ROUND(
100.0 *
SUM(CASE
WHEN order_date = customer_pref_delivery_date

THEN 1
ELSE 0
END)
/ COUNT(*),
2
) AS immediate_percentage
FROM FirstOrders
WHERE rn = 1;
```

Case 1

Input

Delivery =

delivery_id	customer_id	order_date	customer_pref_delivery_date
1	1	2019-08-01	2019-08-02
2	2	2019-08-02	2019-08-02
3	1	2019-08-11	2019-08-12
4	3	2019-08-24	2019-08-24
5	3	2019-08-21	2019-08-22
6	2	2019-08-11	2019-08-13

View more

Output

immediate_percentage
50